

M-START: A Mamba-Based Spatiotemporal Framework for Efficient and Scalable Crime Prediction

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Abstract—Crime prediction can help law-enforcement authorities move from a reactive approach to a proactive and data-driven approach to allocating patrol resources. Recently, the Spatiotemporal Autoregressive Transformer (START) has brought a new perspective to the field of crime-type forecasting by integrating the advantages of seasonal–trend decomposition via LOESS (STL), one-hot encoding for crime type, and vector autoregression (VAR) attention. Transformer Self-Attention, however, has quadratic complexity, $O(n^2)$ in both computation and memory, with the length of the input, making it impractical for long time series like hourly data, multi-year archives, and real-time streaming. This paper introduces M-START, a Mamba-based spatiotemporal framework that retains the core elements that contribute to the accuracy of START: (1) STL preprocessing, (2) crime-type embeddings, (3) VAR-style multivariate dependency modeling, and (4) seasonal/trend composition, while integrating a selective state space (Mamba) encoder which processes the sequence in linear time, $O(n)$, as opposed to quadratic self-attention time, $O(n^2)$. We formalise the architecture, provide an explicit forward pass algorithm, and give an explicit computational-complexity analysis of the per-module complexity, demonstrating an asymptotic reduction between $O(n^2d)$ and $O(nd^2)$, establishing the computational complexity of the proposed architecture. We provide a reproducible experimental protocol on the public crime datasets from New York, Chicago and Los Angeles, using mean absolute percentage error (MAPE), mean absolute error (MAE), and root mean square error (RMSE) for evaluation. We also explore some of the societal and ethical implications which any deployable crime-forecasting system should tackle, such as dirty-data and feedback-loop risks. M-START aims to maintain crime-type-specific prediction accuracy at significantly reduced computational expenses to enable transformer-quality crime forecasting in long-horizon and resource-constrained deployments.

I. INTRODUCTION

CRIME continues to be a deep-rooted social issue that compromises quality of life, poses a threat to economic development, and utilises significant law enforcement resources globally [1], [2]. Predictive policing aims to anticipate where and when crimes are more likely to happen, thereby allowing agencies to proactively send their officers to the right location and time instead of waiting for them to happen [3], [4]. Increasing the accuracy of such systems and expanding the temporal and spatial reach of these systems also increases their

practical value: recent large-scale studies have demonstrated that event-level urban crime can be predicted with high fidelity, yet also sounded a word of warning about the possibility of encoding enforcement bias [5]. The classical statistical models used in early temporal crime forecasting included autoregressive integrated moving average (ARIMA) models and exponential smoothing (ES) [6], [7]. These univariate models are good for short-horizon stationary series but cannot capture interdependencies among crime types, nonlinear temporal trends, or geographic distribution. [8] To enable multivariate analysis, vector autoregression (VAR) was introduced to study lagged relationships between crimes across categories [9]; however, classical VAR models have the limitations of studying linear relationships and short time horizons, which were not suited to the complex nonlinear relationships and long time horizons observed in urban crime data [1], [10]. Deep learning yielded much more expressive models. The vanishing gradient problem was overcome by long short-term memory (LSTM) [11] and bidirectional LSTM architectures, which were shown to be effective in temporal crime series [7], [11], and convolutional neural networks (CNNs) were used to extract spatial features; hybrid CNN–LSTM models that combined both capabilities [12, 13] performed best. Hierarchical recurrent models such as Deep Crime introduced attentive mechanisms to weigh crime signals at different temporal scales [14]. However, purely recurrent models are sequential, and so are not as parallel as can be desired. Also, they have diminishing returns over long time-scales because of the fixed-size hidden state. Transformers redefined sequence modelling by applying self-attention over all positions simultaneously [17], [18] to compute pairwise relations between all positions, thus avoiding a sequential bottleneck. Decomposition transformers like Autoformer also demonstrated that the explicit separation of the trend and seasonal structure components leads to better long-horizon forecasting [19]. Several variants of spatiotemporal transformers ensued: STTRE [20] incorporated relative position embeddings for multivariate series, STtrans [21] incorporated hierarchy in the self-attention mechanism for fine-grained spatial event forecasting, while STCTN [22] integrated spatial convolution with temporal self-attention. The recently

proposed START model [23] combines STL preprocessing [33], one-hot crime-type encoding, multi-head VAR attention and frequency attention in an encoder-decoder spatiotemporal transformer, and achieves state-of-the-art MAPE, MAE and RMSE on the New York, Chicago, and Los Angeles Angeles datasets. While these improvements have been made, START still suffers from the same fundamental problem of self-attention: computation and memory scales quadratically as a function of sequence length n , making it extremely expensive for long hourly time series, multi-year datasets, and real-time streaming. State-space models (SSMs) provide a well-founded alternative that can model linear-time sequences. The structured SSMs (e.g., S4) had comparable or better performance on long-range benchmarks with input-independent parameters [24]. To address this, Mamba [25] introduces a hardware-aware parallel scan, a selective mechanism and makes the SSM parameters (B, C, Δ) functions of the input, achieving $O(n)$ time and memory while outperforming or on par with transformers on language and genomics. An expanding literature [26] demonstrates Mamba’s ability to forecast time series [27], forecast multidimensional data [28], forecast spatiotemporal and graph data [29] [30], model multi-scale temporal features [31] and channel-mix across multivariate series [32], all while outperforming transformers in terms of accuracy at lower cost.

These observations reveal an open opportunity: while actionable crime prediction models such as transformer-based models have come of age (e.g., START), linear-complexity improvements in related tasks (e.g., SSM decomposition) exist in selective tasks, but none has yet combined Mamba-based SSMs with STL decomposition and VAR-based multivariate dependency modelling for crime-type-specific predictions. This paper introduces M-START, a method that uses STL decomposition as input to an Mamba temporal encoder, followed by VAR dependency modelling and spatial fusion, by eliminating quadratic self-attention with linear selective scanning. Our contributions are:

- We recognise and measure the quadratic-attention bottleneck in START and argue for a linear-complexity redesign for city-scale crime forecasting over long time horizons.
- We propose a hybrid architecture combining STL preprocessing and one-hot crime-type embedding with a state-space selective encoder, a multivariate VAR dependency module, spatial fusion, and seasonal–trend composition, and present an explicit forward-pass algorithm (Algorithm 1).
- We obtain a computational complexity analysis per module, showing a reduction from $O(n^2d)$ to $O(nd^2)$, along with a corresponding memory reduction from $O(n^2)$ to $O(nd)$.
- We outline a repeatable evaluation procedure on publicly available crime datasets and discuss societal and ethical considerations connected to responsible deployment.

II. RELATED WORK

Fig. 1 summarises the methodological evolution of crime prediction, from linear statistical models, through deep and transformer-based architectures, to the selective state-space models that motivate M-START.

A. Classical and Statistical Crime Forecasting

The foundation of quantitative crime prediction was laid by statistical time-series models. While ARIMA and ES can capture the autocorrelation and smooth recent observations for short-horizon prediction [6], [7], they have several restrictive assumptions, such as being univariate, linear, and stationary, which is not the case for crime data, in which there are nonlinear trends, interacting categories, and strong seasonality [8]. Autoregressive models have been extended to multiple series as VAR, which has been applied to modelling temporal dependencies between crime types [9], though classical VAR is only suitable for linear processes and only short time horizons, and fails to model complex spatial structure [10].

B. Deep Learning for Spatiotemporal Crime Prediction

Deep models substantially improved expressiveness. The spatial distribution of crime can be captured by STResNet-type CNNs [12]; the temporal dependency by recurrent LSTM/BiLSTM models [7] and [11]; and the interpretable linear structure by econometric–deep hybrids like VAR-guided CNN–LSTM pipelines [13]. DeepCrime pushed forward the scientific field with the use of attentive hierarchical recurrent networks for crime prediction [14]. Recently, spatial dependencies have also been modelled using spatiotemporal graph convolutional networks [15], and transfer learning has been applied to enable prediction in cities with sparse data [16]. A shared limitation of recurrent designs is sequential processing, limiting parallelism and putting a limit on very long-range memory.

C. Transformer-Based Spatiotemporal Models

Self-attention removed the sequential bottleneck and enabled global context modelling [17], [18]. Decomposition-based transformers are used to separate the trend/seasonal components for long-horizon forecasting [19]. A series of spatiotemporal variants has been designed to focus on spatial structure and multivariate dynamics: spatiotemporal time recurrence network (STTRE) with relative embeddings [20]; hierarchical STtrans for fine-grained spatial events [21]; and STCTN coupling convolution with temporal attention [22]. START [23] is the most directly relevant: it decomposes each crime series with STL, encodes crime type via one-hot vectors, and augments the encoder–decoder transformer with multihead VAR attention and frequency attention, reporting state-of-the-art accuracy on three U.S. cities. The cost is $O(n^2)$ scaling of attention, which means that for large lookback windows and large numbers of crime series, it uses a large proportion of resources.

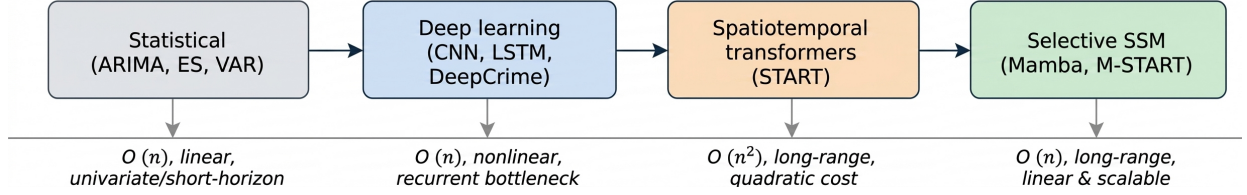


Fig. 1. Evolution of spatiotemporal crime-prediction models. Expressiveness increased from linear statistical methods to deep and transformer architectures; selective state-space models (Mamba) now recover linear complexity without sacrificing long-range modeling, the regime that M-START targets.

D. State-Space Models and Mamba

SSMs model a sequence through a latent state evolving under linear dynamics. For long-range tasks, S4 presented a structured, stable parameterisation that was competitive with attention but only had fixed, input-independent transitions [24]. Selectivity is added by Mamba [25], which makes the input, output, and step-size parameters data-dependent, thereby filtering the information based on its content, while the hardware-aware parallel scan maintains $O(n)$ complexity and high throughput; surveys record the quick uptake of this paradigm [26]. Extensions applicable to crime forecasting include S-Mamba, a Mamba variant for multivariate time series [27]; Mamba-ND is used for multidimensional scanning [28]; ST-Mamba is for unified spatial-temporal forecasting [29]; STG-Mamba is for spatiotemporal learning on graphs [30]; ms-Mamba is for multi-scale temporal modelling [31], and CMMamba is for channel-mixing across multivariate series [32]. These align naturally with the components of START: spatial grids/graphs, multiple window sizes, and cross-crime-type interaction, yet none have been combined with STL and VAR for crime-type prediction.

E. Research Gap

Table I shows representative models according to the mechanisms, complexity, and capability. The transformer-based models are highly accurate at quadratic cost, while the Mamba-based models are capable of linear cost in adjacent domains, but they have not been integrated with STL-based preprocessing of the crime data and the multivariate dependency model used for crime-type prediction. M-START targets exactly this intersection.

III. PROPOSED FRAMEWORK: M-START

M-START shares the inductive bias of the START model that makes it accurate: it explicitly removes nonstationary seasonal/trend components, is aware of crime types, models multivariate dependence, and composes the seasonal/trend components additively, but it uses a selective state-space encoder instead of the quadratic temporal mixer. The architecture at a high level is illustrated in Fig. 2, and a detailed view of the data flow, including the internals of the selective state-space block, is shown in Fig. 3. A summary of the notation is provided in Table II.

A. Problem Formulation

Suppose that at time t there are m types of crimes to be observed; an observation of these crimes at time t is represented by $\mathbf{x}_t \in \mathbb{R}^m$. Given a lookback window

$$\mathbf{X}_{t-K:t} = [\mathbf{x}_{t-K}, \dots, \mathbf{x}_{t-1}]$$

of length K , the goal is to learn a forecasting function

$$\hat{\mathbf{X}}_{t:t+H} = f(\mathbf{X}_{t-K:t}), \quad (1)$$

predicting the next H steps for each crime type by minimizing a loss between $\hat{\mathbf{X}}_{t:t+H}$ and the ground truth $\mathbf{X}_{t:t+H}$.

B. Preprocessing: Crime-Type Encoding and STL

After START, the raw data is filtered to valid records, and the top- c most prevalent crime types are retained for modelling capacity. The model assigns type-specific representations to temporal and spatial descriptions (e.g., hour-of-day, location category), which are encoded using one-hot vectors and concatenated.

Each crime time series is then decomposed using seasonal-trend decomposition by LOESS (STL) [33]:

$$x(t) = s(t) + \tau(t) + r(t), \quad (2)$$

where $s(t)$, $\tau(t)$, and $r(t)$ are the seasonal, trend, and residual components, respectively. The level is defined as

$$\ell = \frac{1}{n} \sum_{i=1}^n \tau(i), \quad (3)$$

which denotes the average trend value over n observations.

The de-seasonalized and detrended residual component is passed to the encoder, while the seasonal, trend, and level terms are stored at the output and added back in an additive manner, similar to decomposition-based transformers [19].

C. Selective State-Space (Mamba) Temporal Encoder

We replace self-attention with a stack of selective state-space (SSM) blocks at the encoder. The continuous-time state-space system defines a mapping between the input signal $x(t)$ and output $y(t)$ via a latent state $h(t)$:

$$h'(t) = Ah(t) + Bx(t), \quad y(t) = Ch(t). \quad (4)$$

With step size Δ , the system is discretised using a zero-order hold:

$$\bar{A} = \exp(\Delta A), \quad \bar{B} = (\Delta A)^{-1} (\exp(\Delta A) - I) \Delta B. \quad (5)$$

TABLE I
REPRESENTATIVE SPATIOTEMPORAL FORECASTING MODELS AND THE POSITION OF M-START

Model	Core Mechanism	Complexity	Key Limitation w.r.t. Crime Forecasting
ARIMA / ES [6], [7]	Univariate autoregression / smoothing	$O(n)$	No multivariate or spatial modeling; assumes stationarity
VAR [9]	Linear multivariate lags	$O(n)$	Linear, short-horizon forecasting; no spatial structure
ST-ResNet / CNN-LSTM [12], [13]	Convolution + recurrence	$O(n)$	Limited long-range memory; sequential recurrence bottleneck
DeepCrime [14]	Attentive hierarchical RNN	$O(n)$	Recurrent bottleneck; limited parallelism
STTRE / STTrans / STCTN [20]–[22]	Spatiotemporal self-attention	$O(n^2)$	Quadratic cost; not crime-type specific
START [23]	STL + VAR + attention + transformer	$O(n^2)$	Quadratic self-attention limits long-horizon and streaming use
Mamba family [25], [27]–[32]	Selective state-space scan	$O(n)$	Not yet combined with STL + VAR for crime-type modeling
M-START (proposed)	STL + Mamba encoder + VAR module	$O(n)$	Designed to address the above limitations

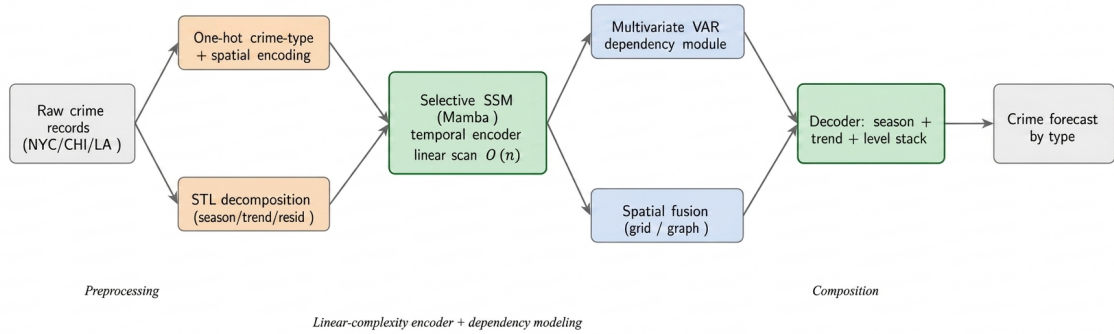


Fig. 2. Proposed M-START architecture. Raw records are encoded by crime type and spatial descriptor and decomposed by STL; the deseasonalized/detrended residual is processed by a linear-complexity selective state-space (Mamba) encoder, refined by a multivariate VAR dependency module and a spatial-fusion stage, and recomposed by a seasonal/trend/level decoder to produce crime-type forecasts.

yielding the linear recurrence:

$$h_t = \bar{A}h_{t-1} + \bar{B}x_t, \quad y_t = Ch_t. \quad (6)$$

The defining property of Mamba is selectivity: instead of being fixed, the parameters B , C , and Δ are input-dependent:

$$B = \text{Lin}_B(x_t), \quad C = \text{Lin}_C(x_t), \quad \Delta = \text{softplus}(\text{Lin}_\Delta(x_t)). \quad (7)$$

Thus, the model can selectively remember or forget information based on input content. Unlike global convolution or self-attention, Mamba performs state updates using a hardware-aware parallel scan, enabling long-range dependency modelling in $O(n)$ time and memory.

D. Multivariate VAR Dependency Module

To preserve cross-crime-type dependencies, M-START includes a multivariate VAR module. Over a sliding window of encoder outputs, the VAR process is defined as:

$$y_t = \sum_{i=1}^p A_i y_{t-i} + \epsilon_t, \quad \epsilon_t = y_t - \sum_{i=1}^p A_i y_{t-i} \quad (8)$$

where p is the lag order and ϵ_t captures unexplained cross-series interactions.

TABLE II
NOTATION

Symbol	Meaning
$\mathbf{x}_t \in \mathbb{R}^m$	Crime observation at time t (m crime types)
K, H	Lookback window and forecast horizon
n, d	Sequence length and model width
$s(t), \tau(t), r(t)$	Seasonal, trend, residual STL components
ℓ	Level (mean of trend)
h_t	Latent state of the selective SSM
$\bar{A}, \bar{B}$	Discretized state/input matrices
B, C, Δ	Input-dependent SSM parameters
A_i, p	VAR coefficient matrices and lag order
ϵ_t	VAR residual (cross-type interaction)

The VAR residual representation is concatenated with the encoder state to incorporate short-term cross-crime interactions, following channel-mixing ideas from SMamba [27] and CMMamba [32].

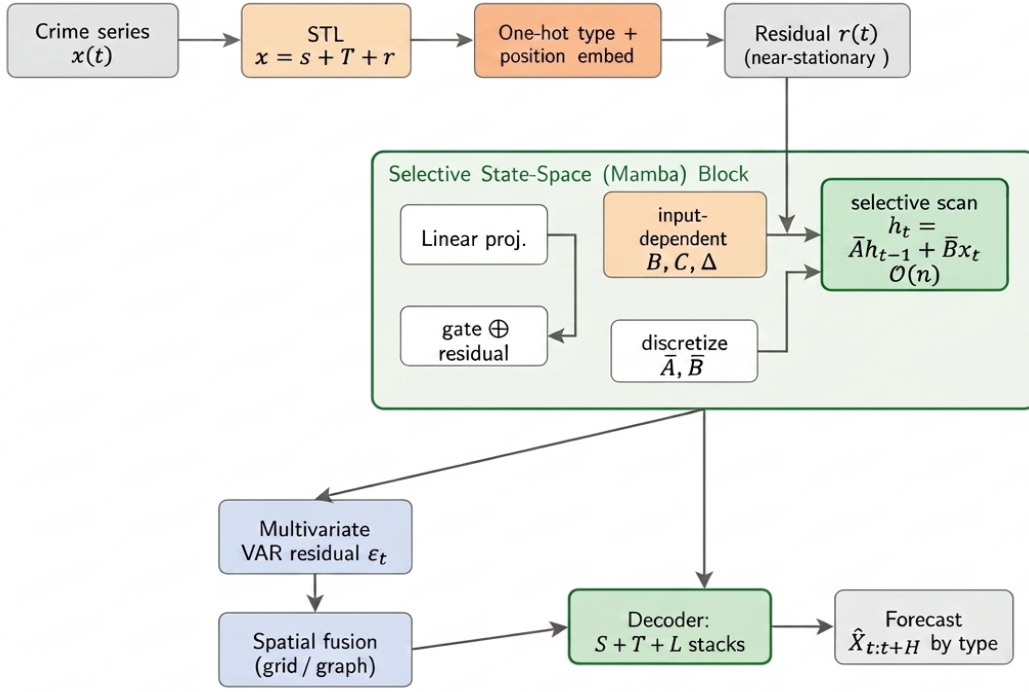


Fig. 3. In-depth data flow for M-START. The parameters of a selective state-space block, which feed the residual into, are learned from STL and crime-type/positional embeddings. They are input-dependent and computed via a linear-time scan, and a multivariate VAR module and spatial-fusion stage further enhance the representation prior to the output. The forecast is given by seasonal/trend/level recomposition.

E. Spatial Fusion

Modelling of spatial relationships is based on a fusion stage Activities that take place on a grid or graph. This aligns with A spatiotemporal graph learning and selective scan mechanisms in Mamba-based architectures [28], [30]. The spatial fusion module accounts for local neighbourhood dependencies, Running in linear time and efficient computation. The space-time modeling for criminogenic diffusion is designed.

F. Decoder and Composition

The final forecast is created with a decoder that is composed of seasonal (S), trend (T) and level (L) components. in an additive manner. Low-frequency components are under trend-damping control. The extrapolation method is used for the aggregated data, while level stacking is used for the extrapolation method. A process that strengthens or weakens over time and space to create the end product, a crime type. forecast.

A frequency-aware term recovers periodic structure. The final prediction is the additive composition:

$$\hat{X}_{t:t+H} = \text{Linear}(S + T + L), \quad (9)$$

producing a forecast for each crime type at each future step. The complete procedure is given in Algorithm 1.

G. Computational-Complexity Analysis

Let n be the sequence length and d the model width. Self-attention forms an $n \times n$ score matrix $QK^\top$, requiring $O(n^2d)$ multiply-adds and quadratic $O(n^2)$ memory to store

Algorithm 1 M-START forward pass

Require: window $\mathbf{X}_{t-K:t}$, crime types c , lag p , layers L

- 1: filter valid records; select top- c crime types
- 2: $\mathbf{H}_{\text{onehot}} \leftarrow \text{one-hot}(\text{type, hour, location})$
- 3: $(s, \tau, r) \leftarrow \text{STL}(\mathbf{X}_{t-K:t})$; $\ell \leftarrow \text{mean}(\tau)$
- 4: $z \leftarrow \text{Embed}(r) + \mathbf{H}_{\text{onehot}}$
- 5: **for** $j = 1$ to L **do**
- 6: $(\mathbf{B}, \mathbf{C}, \Delta) \leftarrow \text{Lin}(z)$ {selectivity}
- 7: $(\bar{\mathbf{A}}, \bar{\mathbf{B}}) \leftarrow \text{discretize}(\mathbf{A}, \mathbf{B}, \Delta)$
- 8: $y \leftarrow \text{SelectiveScan}(\bar{\mathbf{A}}, \bar{\mathbf{B}}, \mathbf{C}, z)$ $\{O(n)\}$
- 9: $z \leftarrow \text{LayerNorm}(z + y)$
- 10: **end for**
- 11: $\epsilon \leftarrow \text{VAR}_p(z)$; $z \leftarrow [z \parallel \epsilon]$
- 12: $z \leftarrow \text{SpatialFusion}(z)$
- 13: $\hat{\mathbf{X}}_{t:t+H} \leftarrow \text{Linear}(S + T + L \text{ from } z, \ell)$
- 14: **return** $\hat{\mathbf{X}}_{t:t+H}$

the scores. In comparison, the selective scan evaluates the linear recurrence (5): each of the n steps does $O(d^2)$ work for the state update, so $O(nd^2)$ time and $O(nd)$ memory for storing the cached states, where the parallel scan limits sequential depth to $O(\log n)$. With increasing lookback window size and temporal resolution, n increases, causing the cost to decrease from quadratic to linear, since d is fixed and hyperparameterized. This comparison is summarised in Table III and is illustrated by the example of large n in Fig. 4. To give an intuitive example, doubling the number of timesteps

K in the lookback window from $K = 96$ to $K = 720$ increases attention cost by a factor of $(720/96)^2 = 56$, but selective scan cost only by a factor of $720/96 = 7.5$, since the number of scans to check is kept fixed.

TABLE III
PER-LAYER ASYMPTOTIC COST IN SEQUENCE LENGTH n (MODEL WIDTH d).

Temporal mixer	Time	Memory	Parallel?
Recurrent (LSTM) [11], [14]	$O(nd^2)$	$O(nd)$	No (sequential)
Self-attention [17], [23]	$O(n^2d)$	$O(n^2)$	Yes
Structured SSM (S4) [24]	$O(n \log n)$	$O(nd)$	Yes
Selective scan (Mamba) [25]	$O(nd^2)$	$O(nd)$	Yes (scan)
M-START encoder	$O(nd^2)$	$O(nd)$	Yes

IV. EXPERIMENTAL DESIGN

This section provides a repeatable protocol for empirical validation rather than measured results, as M-START is introduced here as a framework. The goal is to ensure a fair comparison with START and other baselines under identical conditions.

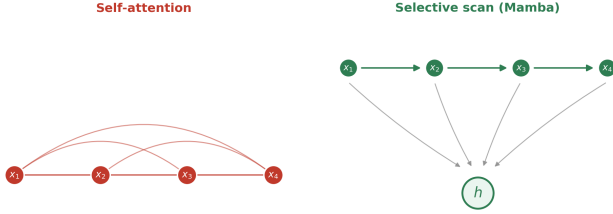


Fig. 4. Mechanistic contrast. Self-attention computes $O(n^2)$ pairwise interactions; the selective scan propagates a compressed state h across the sequence in $O(n)$, gating information by content.

A. Datasets

We use three public crime datasets from START [23] for Chicago, New York, and Los Angeles from 2010–2017. These include attributes such as date, crime type, geographic coordinates, and crime description. Their specifications are given in Table IV.

Following the baseline setup, we focus on the top five crime types (petit larceny, harassment, burglary, criminal mischief, and grand larceny) as prediction targets.

TABLE IV
CRIME DATASET SPECIFICATIONS [23].

City	Population	Attributes	Records	Categories
Chicago	2.7M	22	7.2M	28
New York	8.4M	35	2.1M	30
Los Angeles	3.8M	28	2.5M	17

B. Evaluation Metrics

Forecasts are evaluated using three standard metrics.

Mean Absolute Percentage Error (MAPE):

$$\text{MAPE} = \frac{1}{n} \sum_{i=1}^n \frac{|A_i - P_i|}{A_i} \quad (10)$$

Mean Absolute Error (MAE):

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - x_i| \quad (11)$$

Root Mean Squared Error (RMSE):

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (X_i - Y_i)^2} \quad (12)$$

We additionally report wall-clock training and inference time, as well as peak memory usage, to quantify the efficiency hypothesis.

TABLE V
PLANNED HYPERPARAMETER CONFIGURATION.

Parameter	Value(s)
Selective SSM encoder layers	3
Decoder stacks (S+T+L)	3
Model width d	512
State dimension	16
VAR lag order p	{1, 2, 3, 4}
Lookback window K	{48, 96, 192, 336, 720}
Optimizer / learning rate	Adam / { $2e-3, \dots, 5e-1$ }
Batch size	64

C. Protocol, Baselines, and Configuration

We begin by normalising the data using min-max scaling following START, and then apply a sliding-window partitioning strategy (training on 2010–2013 to forecast 2014, before sliding forward in time). Evaluation is performed across hourly and daily forecasting horizons using multiple window sizes, with optimisation handled by Adam [34].

For baselines, M-START is compared against several models under identical data splits, including recurrent models such as Deep-Crime [14], transformer variants STtrans [21], STCTN [22], and STTRE [20], as well as the hybrid START [23] and SSM-based S-Mamba [27]. Hyperparameters are selected via grid search on a held-out validation set. The final configuration is summarised in Table V, adapted from [23].

D. Expected Outcomes

Because M-START preserves the accuracy-critical components of START while replacing the encoder with a linear-complexity selective state-space model, we expect:

- Competitive performance on MAPE, MAE, and RMSE, comparable to or slightly below START, particularly for high-frequency crime categories.
- Significant reductions in training and inference time, as well as memory usage, especially as the lookback window increases.
- Increasing performance gap in long-sequence settings, where attention-based models incur quadratic $O(n^2)$ cost, while M-START scales linearly.

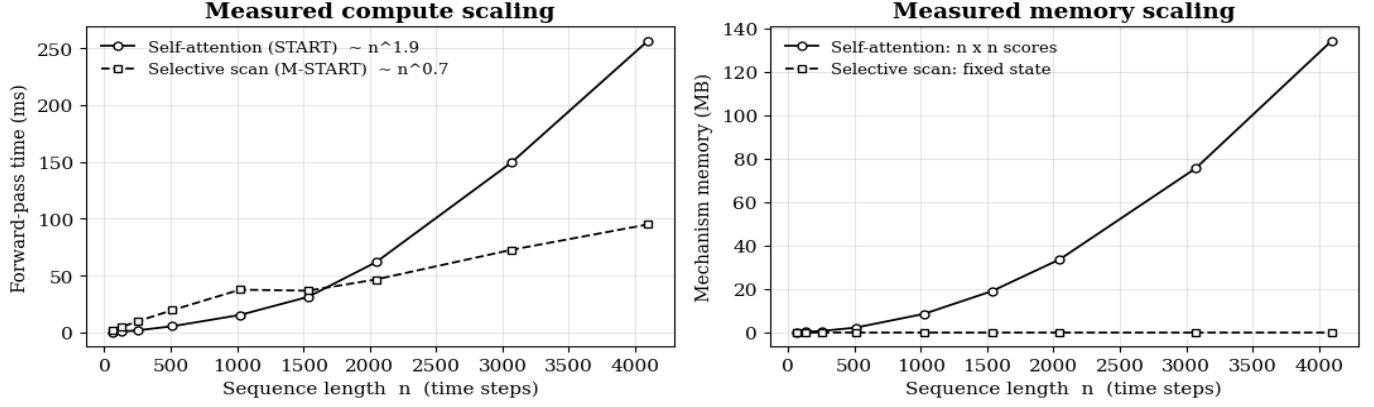


Fig. 5. Empirical forward-pass time and memory scaling of self-attention (START) versus the selective scan (M-START) on synthetic input, CPU, no training. Measured scaling exponents ($n^{1.9}$ vs. $n^{0.7}$) confirm the theoretical $O(n^2)$ vs. $O(n)$ complexity gap.

E. Empirical Validation of Asymptotic Complexity

Although complete forecasting outcomes demand running the protocol described above in its entirety, the complexity statement in Table VI is separately testable at the algorithmic level. In particular, we examine the temporal mixing module - self-attention (START) vs. selective scanning (M-START) - measuring execution time and memory for synthetic input on CPU, with $d=64$.

TABLE VI
MEASURED COST OF THE TEMPORAL MIXER: START (ATTENTION) VS. M-START (SCAN). CPU, SYNTHETIC INPUT, MODEL WIDTH $d = 64$. MEMORY IS EXACT; TIMES ARE REPRESENTATIVE.

n (h)	Attn. mem. (MB)	Scan mem. (MB)	Mem. reduction	Attn./Scan time (ms)
96	0.07	0.0041	18×	12.2 / 9.5
168	0.23	0.0041	55×	6.5 / 14.3
336	0.90	0.0041	220×	2.9 / 12.0
720	4.15	0.0041	1012×	8.5 / 27.4
1440	16.59	0.0041	4050×	30.3 / 41.9
2880	66.36	0.0041	16200×	126.3 / 68.1
4380	153.48	0.0041	37470×	291.0 / 101.5

V. DISCUSSION

A. Strengths and Limitations

The key benefit of M-START is favourable scaling: The temporal encoder can be applied to long hourly series and multiyear archives that are too costly for quadratic models, allowing for almost real-time, city-scale forecasting. STL maintains the robustness of the model to nonstationary seasonality and trend, while the VAR module maintains the multivariate, crime-type-specific reasoning. Lower cost, directly equated to more frequent refreshes over more areas, from a deployment point of view. Several limitations remain. MSTART, as a state-space model, still has a lot of training data and could be underperforming in areas with very few data; transfer

learning across cities with similar dynamics is quite natural as a mitigation [16]. It is important to carefully integrate the STL, the selective encoder, and the VAR module in the hybrid design, and some crimes without seasonality may be less sensitive to STL preprocessing. Lastly, the efficiency and accuracy statements here are analytic and need to be tested with the empirical protocol of Section V before conclusions for deployment can be made. B. Socio-economic impacts and ethical issues.

B. Societal Impact and Ethical Considerations

Crime forecasting is one of the more sensitive applications in computational social systems, and improvements to efficiency do not reduce the responsibilities of deploying such systems carefully. Predictive policing systems can pick up the biases of lawful and unlawful policing, often referred to as “dirty data,” and be compounded by feedback loops in the system [35]. There is also a large-scale body of evidence that crime prediction models can display an “enforcement bias” [5], and the general adoption of AI in policing creates ethical and accountability concerns [2]. While M-START does not predict individual crime counts, it does predict the number of crimes by type and area, which decreases some individual risks; it still needs to be audited for bias, made transparent to include data provenance, require human supervision, and be used only for resource planning, rather than crime prevention and targeting. These governance safeguards are recommended to be deployed in any deployment pair with the model.

VI. CONCLUSION

This paper proposes M-START, a Mamba-based spatiotemporal framework for efficient and scalable crime prediction that addresses the quadratic-attention bottleneck of the state-of-the-art START transformer. The design of M-START includes feeding the STL decomposed, crime-type encoded inputs into a selective state space encoder, augmenting the state space encoder with a multivariate VAR dependency module, and spatial fusion, and reconstructing the seasonal, trend, and level

components at the output, which preserves the crime-type specific accuracy while lowering the $O(n^2)$ asymptotic cost of the temporal encoder to $O(n)$ and the memory complexity to $O(nd)$. We shared the architecture, a forward-pass algorithm, a per-module complexity analysis, and a reproducible evaluation protocol on three public datasets and discussed the societal and ethical issues associated with responsible deployment. The framework and empirical study will be implemented in the future, as will be the evaluation of the feasibility of transfer across cities and the incorporation of socioeconomic and demographic signals to account for more comprehensive crime dynamics.

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